



Computer aided diagnosis for suspect keratoconus detection

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ABSTRACT

Purpose: To develop a stable and low-cost computer aided diagnosis (CAD) system for early keratoconus detection for clinical use.

Methods: The CAD combines a custom-made mathematical model, a feedforward neural network (FFN) and a Grossberg-Runge Kutta architecture to detect clinical and suspect keratoconus. It was applied to retrospective data of 851 subjects for whom corneal elevation and thickness data was available. These data were divided into four groups: a control group (312 eyes) with bilateral normal tomography, keratoconus suspect (77 eyes) with a clinically diagnosed keratoconus in one eye and a normal fellow eye, mild keratoconus (220 eyes), and moderate keratoconus (229 eyes). The proposed framework is validated using 10-cross-validation, holdout validation and ROC curves.

Results: The CAD detects suspect keratoconus with an accuracy of 96.56% (sensitivity 97.78%, specificity 95.56%) versus an accuracy of 89.00% (sensitivity 83.00%, specificity 95.00%) for Belin/Ambrosio Deviation (BADD), and an accuracy of 79.00% (sensitivity 58.00%, specificity 99.70%) for Topographical Keratoconus Classification (TKC). For the detection of mild to moderate keratoconus CAD shows nearly similar accuracies as previously described methods, with an average accuracy of 99.50% for CAD, versus 99.46% for BADD and 96.50% for TKC. The proposed algorithm also provides a 70% reduction in computation time, while increasing stability and convergence with respect to traditional machine learning techniques.

Conclusion: The proposed algorithm is highly accurate and provides a stable screening platform to assist ophthalmologists with the early detection of keratoconus. This framework could potentially be set up for any Scheimpflug tomography system.

1. Introduction

Keratoconus (KTC) is a non-infectious pathology that gradually deforms the cornea, the transparent structure at the front of the eye that focuses light through the lens on the retina. Healthy cornea has a convex aspherical shape and a large difference in refractive index compared to air, which makes it the strongest refractive element in the eye. Thus, any corneal alterations or irregularities, as seen in keratoconus, cause major vision impairment [1,2]. Typical signs of KTC range from progressive nearsightedness, irregular astigmatism, distorted or blurred vision, difficulties with night driving, and light sensitivity [3,4]. The incidence and prevalence of keratoconus vary depending on factors such as gender and ethnicity [3]. Nevertheless, keratoconus epidemiology is still unclear. An analysis that covered the entire Danish

population reported an estimated prevalence of 86 keratoconus patients per 100 000 residents [5], while a more recent study suggests that the prevalence is significantly higher than around 1:2000 [6]. The condition usually affects both eyes, albeit not necessarily simultaneously. Although the underlying causes are not well understood, there are indications that the disease is multifactorial in origin, related to genetic predisposition and environmental factors such eye rubbing, certain eye conditions and family history [1–3].

In clinical practice the highly irregular shape of advanced keratoconic cornea is easily identified, while the detection of early stages of the disease remains an important challenge. Therefore, developing an efficient technique to assist practitioners with objectively detecting early keratoconus is of paramount importance, as it allows planning more effective treatment strategies and avoiding further acuity loss.

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Table 1
Description of the study groups.

	K_{mean} (D)	K_{max} (D)	$Pachy_{min}$ (μm)	Astig (D)	I–S value
Control (312 eyes)	43.28 $\pm$ 1.29 (39.30, 46.60)	45.11 $\pm$ 1.56 (40.65, 47.80)	547.06 $\pm$ 30.88 (496, 615)	0.86 $\pm$ 0.39 (0.2, 1.9)	0.20 $\pm$ 0.60 (-1.49, -1.33)
Suspect KTC (90 eyes)	42.89 $\pm$ 1.51 (39.60, 46.10)	44.03 $\pm$ 1.50 (41.23, 47.71)	530.50 $\pm$ 34.37 (458, 591)	0.77 $\pm$ 0.48 (0.1–2.2)	0.64 $\pm$ 0.47 (-0.27, -1.61)
Mild KTC (220 eyes)	43.89 $\pm$ 2.21 (39.10, 60.80)	51.77 $\pm$ 3.70 (43.40, 67.90)	477.60 $\pm$ 34.94 (411, 565)	2.98 $\pm$ 1.58 (0.10, 7.60)	4.74 $\pm$ 1.56 (0.7, 8.3)
Moderate KTC (229 eyes)	45.32 $\pm$ 3.68 (39.80, 63.60)	58.68 $\pm$ 6.03 (49.60, 81.40)	449.26 $\pm$ 39.28 (306, 528)	4.01 $\pm$ 2.12 (0.2, 12.6)	9.50 $\pm$ 2.97 (1.67, 19.62)

Mean $\pm$ SD (range).

I–S, inferior-superior value; KTC, keratoconus; K_{max} , maximum keratometry; $Pachy_{min}$, minimum pachymetry; Astig, anterior corneal astigmatism.

Keratoconus diagnosis was initially addressed by the introduction of videokeratography and later by Scheimpflug tomography. Videokeratography provides accurate information about anterior corneal surface shape, while the Scheimpflug tomographer adds information on the posterior corneal surface and corneal thickness. The availability of these measurements led to the introduction of various indices for keratoconus detection. Several authors developed keratoconus detection approaches based on a combination of parameters, such as, keratoconus index (KCI%) [7], inferior superior value (I–S value) [8], keratoconus severity index (KSI) [9], KISA% index [10], keratoconus severity score (KSS) [11], Score [12], Belin/Ambrosio Enhanced Ectasia Display (BAD-D) [13], topographical keratoconus classification (TKC), and more recently Pentacam Random Forest Index (PRFI) [13–33]. Most of these approaches introduce a specific combination of parameters to machine learning algorithms.

Machine learning (ML) algorithms are rapidly gaining importance in the health sector. They are often used to assist in medical diagnosis tasks by capturing the most relevant clinical features of complex data, providing early interpretable results, and helping to choose suitable treatment strategies [34–37]. In the past two decades, several methods have been proposed to automatically detect keratoconus using mathematical models and machine learning [3,22,23,38–44]. These efforts were limited as they only considered ‘classic’ forms, rather than hybrid approaches of modelling and machine learning. Studies that focused on keratoconus detection employed traditional machine learning to corneal elevation or other extracted clinical parameters, thus neglecting practical restrictions, such as the high variability and sometimes the unstructured nature of corneal tomography data, which requires long computational time. Moreover, the classic algorithms are sensitive to the initial condition and show low accuracies in the earliest cases of the condition. The instability within the earliest cases reduce the performance and the practical use of ML algorithms. However, more stable results are usually achieved in the absence of highly correlated groups, such as advanced keratoconus versus normal or after applying appropriate data preprocessing techniques [45,46]. Some previous works therefore suggested the use of mathematical modelling followed by statistical analysis of clinically derived parameters [28,29,47,48]. These studies are potentially low cost, but mostly platform dependent as they often use proprietary parameters associated with one particular commercial device or manufacturer, thus hampering cross-platform reproducibility of the results.

This paper proposes a computer aided diagnosis (CAD) system for the automatic detection of keratoconus, based on corneal tomography maps and minimum corneal thickness that can be captured by any Scheimpflug device. This system takes the highly unstructured and uninterpretable number of datasets into consideration, as well as the algorithm's instability to the earliest cases of the disease, while also improving the accuracy. The system consists of three main steps. First, a custom-made mathematical model of keratoconic cornea is presented to transform the unstructured topographies to a limited number of meaningful features. Next, a feedforward neural network (FFN) is used to process the model derived features. This type of neural network was selected since it performs better when dealing with corneal topographies [38]. To preserve the stability and improve the accuracies in detecting KTC, a Grossberg architecture was used based on an iterative

numerical method, combined with both the feedforward neural network and the mathematical model.

The proposed system overcomes the inherent complexity of tomographical data and permits an accurate detection of keratoconus at the early stage. This framework also optimizes the memory space required in comparison to the direct use of feedforward neural network [38]. Moreover, the CAD has the potential to be platform independent and reproducible on other systems since it only relies on corneal tomography and pachymetry, which is available in all Scheimpflug systems. This requires further validation, however. Finally, the principles of proposed iterative method can be applied to improve the stability and the repeatability of other machine learning techniques.

2. Materials and methods

2.1. Patient data

This work uses a previously collected database containing the Pentacam (Oculus GmbH, Wetzlar, Germany) Scheimpflug measurements of 851 subjects (aged 33.9 ± 9.5 years). Of these subjects 312 presented bilateral normal topography, with no systemic or ocular disease, without any prior ocular surgery, and no slit lamp findings suggestive of cornea ectasia. These normal subjects were recruited during a previous epidemiology study [49]. Next, there were 539 keratoconic subjects divided into three groups. The first group contains mild keratoconus (220 eyes), defined as having a clear cornea, tomography maps compatible with keratoconus (for parameter ranges see Table 1), a Fleischer ring at the apex base, slight thinning, and anterior and/or posterior corneal steepening. Next, there is the moderate keratoconus group, which included 229 subjects that had corneas with slit-lamp findings compatible with keratoconus (see parameters in Table 1), corneal thinning at the apex, Vogt striae, a clearly visible Fleischer ring and corneal tomography compatible with keratoconus. Finally, there was a keratoconus suspect group (90 eyes), defined as unilateral keratoconus, with one eye presenting a non-symptomatic cornea with normal tomography and no biomicroscopic signs, while the other cornea of the other eye has clinically diagnosed ectasia [24,50]. For each group only, right eyes were considered. A description of the groups is given in the following (Table 1).

All measurements were screened by both an ophthalmologist and an optometrist for quality and validity. Subjects with a self-reported history of ocular surgery, systemic or ocular disease were excluded. The data required for analysis consists of the anterior, posterior corneal elevation for a range of ± 7 mm around the corneal apex in 0.1 mm steps and minimum corneal thickness. The data were created by exporting elevation maps to Pentacam CSV files, which were imported in Matlab (R2017a) in the form of 141×141 anterior and posterior matrices, $Z_A(x,y)$ and $Z_P(x,y)$ respectively. The study was approved by the Antwerp University Hospital (UZA, Belgium) Ethical Committee and adhered to the tenets of Declaration of Helsinki. All healthy subjects provided written informed consent before inclusion. Meanwhile, the keratoconic subjects were included retrospectively and required no informed consent according to Belgian law.

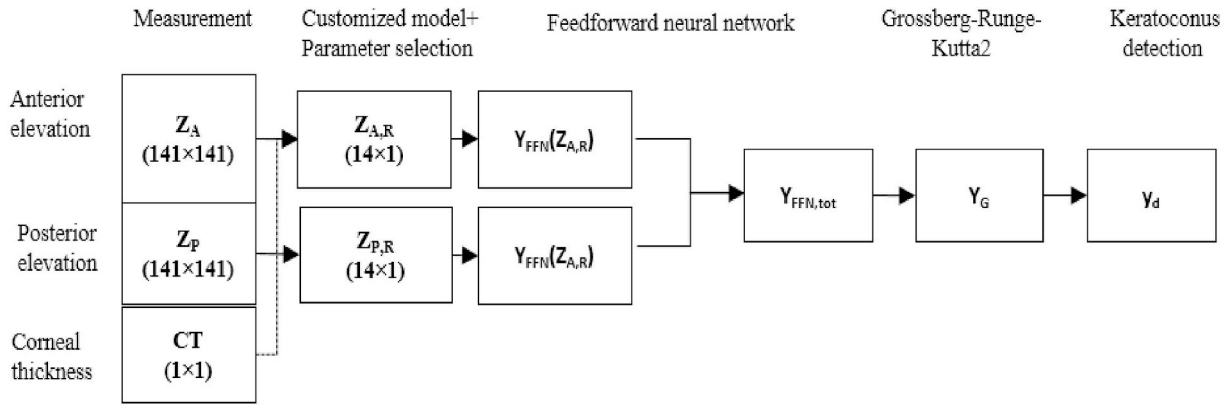


Fig. 1. Overview of the modelling and data processing used in the CAD system.

2.2. Mathematical model

The proposed framework starts from a customized mathematical model (CMM) to reduce the complexity and dimensionality of the topographical data by transforming each non-meaningful, unstructured matrix Z_A and Z_P to an interpretable vector. This reduces execution time and memory requirements, while improving the performance of the machine learning algorithm. Several mathematical models of the cornea [39,40,49,51–56] can be used to transform and reduce the matrices Z_A and Z_P . Previously, Navarro et al. introduced a mathematical decomposition of anterior normal corneal surface by combining a quadric function plus an irregular component [53]. Following the same idea Martínez-Finkelshtein et al. developed an iterative algorithm that dynamically adds terms to a sphere basis function [54], thus enabling a proper modelling of corneal irregularities. By following the same approach, and given the general shape of keratoconic cornea, we opted to use a sum of an ellipsoid surface as a basis function plus a 2-D Gaussian function:

$$z(x, y) = f_s(x, y) + f_G(x, y) \quad (1)$$

where, z is the elevation, f_G the Gaussian function and f_s the ellipsoid function. The z -axis corresponds with the ellipsoid's optical axis, while $z(x, y)$ corresponds with the distance to a reference plane $z = 0$.

The ellipsoid function is given by:

$$f_s(x, y) = C_x \cdot (x - x_0)^2 + C_y \cdot (y - y_0)^2 - cst \quad (2)$$

with (C_x, C_y) the curvature of the ellipsoid, (x_0, y_0) the ellipsoid center, and cst a constant. Meanwhile, the Gaussian function is given by:

$$f_G = A \cdot \exp \left[-\frac{x_{rot}^2}{2\sigma_x^2} - \frac{y_{rot}^2}{2\sigma_y^2} \right] \quad (3)$$

where A is the Gaussian amplitude and (σ_x, σ_y) the standard deviations. This function uses rotated coordinates (x_{rot}, y_{rot}) to allow for keratoconus orientations other than the x - or y -directions and are given by:

$$\begin{cases} x_{rot} = (x - \mu_x) \cdot \cos\theta - (y - \mu_y) \cdot \sin\theta \\ y_{rot} = (x - \mu_x) \cdot \sin\theta + (y - \mu_y) \cdot \cos\theta \end{cases} \quad (4)$$

with θ the rotation angle and (μ_x, μ_y) is the Gaussian center. This model assumes that a normal cornea can be represented by a sum of an ellipsoid with a Gaussian term centered around the minimum corneal thickness, to account for the local irregularities found around the thinnest point of a keratoconic cornea. It is therefore expected that the Gaussian component will behave differently in normal corneas than in keratoconic cases (e.g. centered at the corneal apex and very wide for normal corneas, vs. narrow and inferiorly decentered for keratoconus).

By independently fitting the anterior and posterior corneal elevation Z_A and Z_P to equation (1) the model can be customized to each of the

312 normal and 539 keratoconic subjects. From these fits a set of 17 candidate features was derived to identify keratoconus, which could be divided into three groups: ellipsoid parameters (C_x, C_y, x_0, y_0, cst), Gaussian parameters ($A, \sigma_x, \sigma_y, \mu_x, \mu_y, \theta$), and geometric features; ellipsoid volume S_V , ellipsoid surface S_S , Gaussian volume G_V , Gaussian surface G_S , Euclidian norm distance between the centers of the anterior and posterior ellipsoid fits, Euclidian norm distance between the centers of the Gaussian and ellipsoid fits. Finally, the minimal corneal thickness (CT), as well as its coordinates (x_{min}, y_{min}) were added to the analysis, which are known to correlate well with keratoconus [1,2].

These 17 anterior and 17 posterior candidate features in addition to CT and its coordinates were subsequently analyzed for their suitability to discriminate between normal and keratoconic corneas using Neighborhood Component Analysis for Features Selection (NCAFS), which is one of the most efficient techniques for dimensionality reduction [57,58]. The test showed that most parameters perform well in keratoconus detection, except for $C_x, C_y, \theta, G_V, cst$, and x_{min} . Consequently, Z_A and Z_P were reduced to a set of 14 anterior and 14 posterior parameters ($Z_{A,R}$ and $Z_{P,R}$) that may be used to identify keratoconus (Fig. 1). All the analysis was performed in Matlab.

2.3. Computer aided diagnosis

2.3.1. Standard feedforward network applied to tomography data

Feedforward neural network (FFN) is a type of artificial neural network, inspired by biological neurons, that can self-adapt, self-organize and learn from examples, much like their biological counterparts. It is a supervised algorithm that adjusts an input signal X to the desired output Y_d according to an optimization learning rule [59]. The algorithm involves a forward-propagating step followed by a backward-propagating step and it is typically composed of three layers: an input layer that receives m input signals, a hidden layer of n processing units (neurons) with sigmoid activation function f_1 , and an output layer with linear activation function f_2 . Several optimization algorithms can be used to train the neural network, of which backpropagation and Levenberg-Marquardt are popular choices. In this work, the FFN was configured in Matlab, based on a single hidden layer with 15 neurons, a Logsigmoid activation function, a Levenberg-Marquardt training algorithm, and a learning rate of 0.001: With $f_1(t) = \frac{1}{1 + \exp(-t)}$ and $f_2(t) = t$.

The reduced vectors $Z_{A,R}$ and $Z_{P,R}$ for each subject were considered as network's input, while the output was 0 for normal and 1 for any stage of keratoconus (suspect, mild, moderate). This classification is performed independently for each stage as normal versus suspect KTC, normal versus mild KTC and normal versus moderate KTC.

The structure of the standard feedforward network employed is illustrated in (Fig. 2). The standard approach, based on analyzing a large amount of unstructured elevation data, often leads to overfitting, instability and long training times (in the order of minutes). Using a

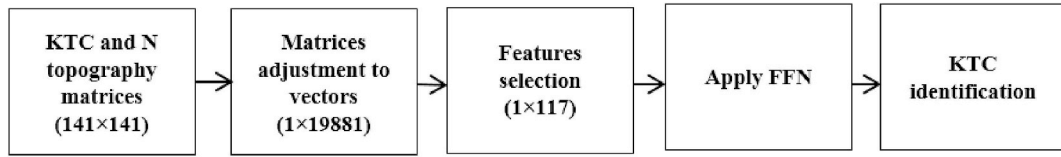


Fig. 2. Classical design of the standard FFN applied to topographical data.

mathematical model combined with Grossberg-Runge Kutta architecture may improve the capabilities of FFN, as presented in detail in the next section.

2.3.2. Grossberg-Runge Kutta architecture

Grossberg architecture is a competitive type of neural network in which a supervised learning process coordinates the weights of its connections according to a desired output. The convergence of Grossberg algorithm to a single state was already demonstrated in the literature [60]. The learning process is based on a simple explicit rule for weights adjustment between inputs and outputs and may be regarded as a mathematical Cauchy problem (eq. (5)) that admits a unique solution under certain assumptions. Several iterative numerical methods have been suggested to approximate the solution $w_{ij}(t)$ of problems (eq. (5)) [61], often referred to as being either *explicit* or *implicit*. These methods are based on integration and discretization processes that lead to different levels of precision, convergence, stability and computational cost. Runge Kutta is one of the best known of these techniques [62–64]. In our case, the Cauchy problem is given by:

$$\begin{cases} \frac{dw_{ij}}{dt} = f(t, w_{ij}(t)) \\ w(t=0) = \text{given} \end{cases} \quad (5)$$

Where w_{ij} is the weight matrix of the Grossberg network and f is the training rule.

Here, the explicit second order Runge Kutta (RK2) was suggested as a learning rule of the Grossberg layer, which is supposed to lead to better convergence and stability compared to the traditionally used learning process. This explicit rule of Grossberg architecture based RK2 is:

$$\begin{cases} v_{ij}(t + \Delta t) = v_{ij}(t) + \delta \cdot (y_i - v_{ij}(t)) \cdot z_i \\ w_{ij}(t + \Delta t) = w_{ij}(t) + \frac{\Delta t}{2} \cdot [\delta \cdot (y_i - v_{ij}(t)) \cdot z_i + \delta \cdot (y_i - v_{ij}(t + 1)) \cdot z_i] \\ w_0 \text{ initial condition} \end{cases} \quad (6)$$

Where, v_{ij} is an intermediate neuronal connection, z_i is the feedforward network output, Δt is a time step, δ is the learning rate of Grossberg network, w_0 is the initial weight. The Grossberg-RK2 was implemented in Matlab with $\delta < 1$ and $\Delta t = 1$, the initial weights were generated randomly between $[0,1]$.

2.3.3. Hybrid machine learning algorithm (CMM-FFN)

The hybrid machine learning algorithm was implemented by combining the custom mathematical model with an adapted FFN according to a probabilistic function (eq. (7)). In practice the FFN with the previous standard configurations (section 2.3.1) were trained on the reduced anterior data $Z_{A,R}$ and the reduced posterior data $Z_{P,R}$, separately, yielding responses y_{Ante} and y_{Post} . These responses were then inserted into the probability function (eq. (7)) to give more weight to the posterior layer, considering that keratoconus often appears on the posterior surface first [65]. This yields a total response function $y_{FFN,tot}$ (Fig. 1), defined as follows:

$$y_{FFN,tot} = p_1 y_{Ante}(Z_{A,R}) + p_2 y_{Post}(Z_{P,R}) \text{ with } p_2 > p_1 \quad (7)$$

During the entire training process all weights are optimized to ensure that the algorithm response $y_{FFN,tot}$ agrees with the original training data y_d as much as possible. Beforehand, weight parameters p_1

and p_2 are estimated according to the previous domination of the posterior surface to approach the desired response as follow:

$$\text{Maxf}(p_1, p_2) = p_1 \cdot y_{Ante} + p_2 \cdot y_{Post} \quad (8)$$

$p_1 < p_2 \in [0,1]$

The computer aided diagnosis system was obtained by combining the CMM-FFN with Grossberg network, therefore, the final output $y_{FFN,tot}$ is given to the Grossberg layer to correct the weights and improve the identification process.

The computer aided design algorithm is then summarized, by the following steps:

- 1) Fit $Z_A(x,y)$ and $Z_P(x,y)$ to the custom mathematical model CMM equation (1).
- 2) Derive and reduce CMM features to $Z_{A,R}$ and $Z_{P,R}$.
- 3) Train FFN-based on [Inputs, Outputs] = $[Z_{A,R}, y_d]$ and compute the algorithm's output y_{Ante} (Anterior).
- 4) Train FFN-based on [Inputs, Outputs] = $[Z_{P,R}, y_d]$ and compute the algorithm's output y_{Post} (Posterior).
- 5) Estimate p_1, p_2 according to: $\text{Maxf}(p_1, p_2) = p_1 \cdot y_{Ante} + p_2 \cdot y_{Post}$
 $p_1 < p_2 \in [0,1]$
- 6) Compute the final output according to:
 $y_{FFN,tot} = p_1 y_{Ante}(Z_{A,R}) + p_2 y_{Post}(Z_{P,R})$
- 7) Implement the Grossberg Layer for the input output data $[y_{FFN,tot}, y_d]$
- 8) Adjust the weights of the Grossberg Layer based on RK2 numerical schemes equation (6).
- 9) Compute the finale output $Y_G = < W, y_{FFN,tot} >$

2.3.4. Validation

To evaluate the performance of the proposed CAD for keratoconus detection, 10-fold cross validation technique was used. This uses data more efficiently than traditional holdout validation. The latter randomly splits the dataset into training sets (70%) and validates the classifier with the other 30%. 10-fold cross validation splits the datasets into 10 independent groups, 9 of which are used for training, while the last group is used for validation. This process is performed a total of 10 times, until each group has been used independently as a test set. This technique better indicates classifier performance, while using new data and providing a more accurate estimation of the error rate [66]. For each of the 10 repetitions the accuracy, precision, sensitivity, and specificity were computed, as defined in (Table 2) [67]. However, to evaluate the general performance of CAD versus CMM-FFN, holdout validation and confusion matrix were used based on the same validation criteria defined in (Table 2).

The final accuracy, sensitivity, specificity, and precision values for 10-fold cross validation were computed as the average of the 10 repetitive times. Both validation techniques (10-fold cross validation, holdout validation) were repeated 20 times and gave roughly the same results each time. Finally, Receiver Operating Characteristic (ROC) curves were used to compare the discriminative abilities of the proposed approach versus CMM-FFN, BADD, and TKC.

3. Results

3.1. Performance of ellipsoid -Gaussian model

The customized model accurately describes the anterior and

Table 2
Definition of validation criteria.

True positive	TP	Number of KTC cases identified correctly
True negative	TN	Number of N cases identified correctly
False positive	FN	Number of KTC cases classified as normal
False negative	FP	Number of normal classified as KTC
Accuracy	$(TP + TN)/(TP + FP + FN + TN)$	Percentage of dataset classified correctly
Sensitivity	$TP/(TP + FN)$	Percentage of KTC cases correctly classified.
Specificity	$TN/(TN + FP)$	Percentage of normal cases correctly classified.
Precision	$TP/(TP + FP)$	The ratio of the correctly classified KTC, high precision relates to low false positive.
Cut-off		Value or point designed as a limit of a group.

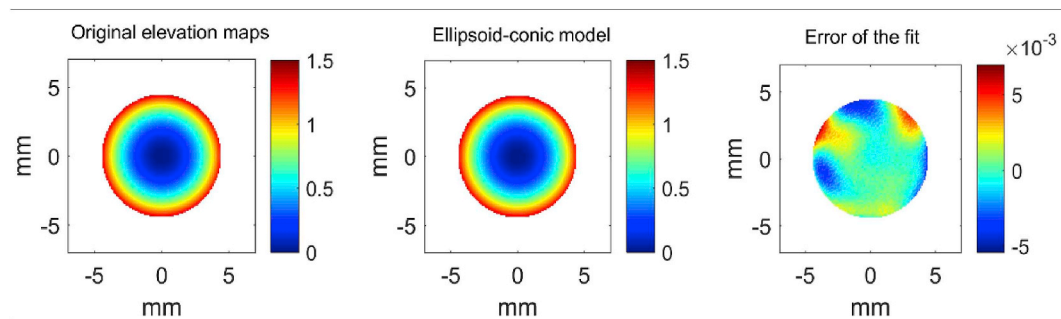


Fig. 3. Representative, randomly chosen example of original elevation topography vs. CMM topography, color bars expressed in mm. Also, x and y axis are expressed in mm. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

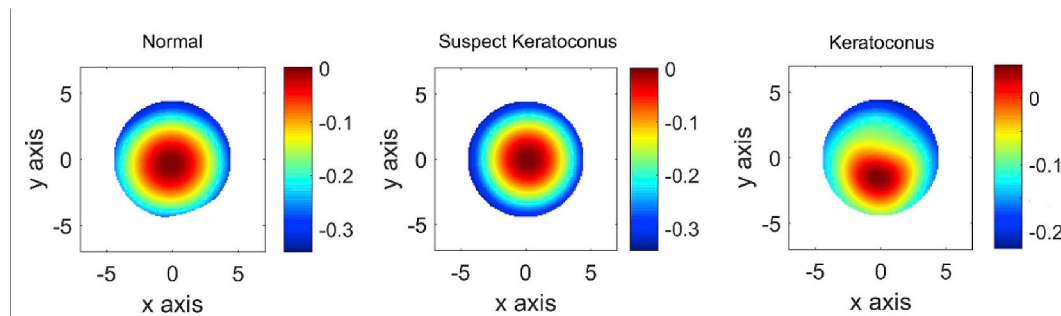


Fig. 4. Representative, randomly chosen examples of the Gaussian term in normal, keratoconus suspect, and moderate keratoconus, color bars expressed in mm. Also, x and y axis are expressed in mm. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

posterior surfaces of both keratoconic and normal corneas. In the center (± 4 mm from the apex) of anterior surface, the best observed error was $\xi \in O(10^{-3})$ mm, while it was $\xi \in O(10^{-2})$ mm at the periphery (radius 4 mm–5 mm). The same best error of $\xi \in O(10^{-3})$ mm was achieved in the center and the periphery of posterior surface (Fig. 3). The model allows for a clear topographic distinction between clinical keratoconic and normal corneas since the Gaussian of normal corneas forms a small, near-perfect circle with a center very close to the apex in the x direction (i.e. $\mu_x \approx 0$) and within ± 2 mm from the apex in the y direction (i.e. $\mu_y \in [-2, 2]$; (Fig. 4)). In most moderate keratoconus cases the Gaussian narrows and undergoes an inferior shift towards $\mu_y \in [-4, -2]$. In suspect keratoconus the CMM was unable to clearly distinguish between early keratoconus and normal (Fig. 4), demonstrating the need for using the machine learning algorithms.

3.2. Performance of the CAD

3.2.1. Influence of sample size

To evaluate the importance of sample size on the results, a balanced set of 312 keratoconus cases, equal to the size of normal group, was randomly selected from the three keratoconus stages (Table 1). Similarly, an imbalanced set of 150 keratoconus was randomly selected from the 312 keratoconus. The validation criteria; accuracy, sensitivity,

specificity, precision and confusion matrix were examined for both groups (balanced and imbalanced). The comparison was given for both CAD and CMM-FFN.

The highest accuracy for the classification of normal versus keratoconus during the validation step for the balanced groups was 98.4% for CAD versus 96.4% for CMM-FFN (Table 3). Since the size of groups might influence the results, this analysis was repeated by training the algorithm using the imbalanced groups of normal and keratoconus, yielding nearly identical results as before for both CAD and CMM-FFN, and show a lower average of change in the sensitivity of CMM-FFN, however, the accuracy and the sensitivity of CAD were higher than the accuracy and the sensitivity of CMM-FFN (Table 3). The results demonstrate the high performance and the general stability of CAD comparing with CMM-FFN when applied to imbalanced data sets.

The normal cases were nearly all classified correctly, with 9 misclassifications of keratoconus as normal for computer aided diagnosis system (Table 4) against 21 misclassifications for the feedforward neural network with the custom mathematical model (Tables 4 and 5). The obtained results therefore indicate that adding Grossberg-RK2 leads to improve the values of the validation criteria.

3.2.2. Comparison between methods

The proposed computer diagnosis system results in nearly identical

Table 3
Performance of CAD vs. CMM-FFN using Holdout Validation.

	CAD				CMM-FFN			
	Accu-racy	Sensi-tivity	Speci-ficity	Preci-sion	Accu-racy	Sensi-tivity	Speci-ficity	Preci-sion
Imbalanced dataset								
Normal (312)	98.2%	96.0%	99.3%	98.6%	96.3%	91.3%	99.3%	98.5%
Keratoconus (150)								
Balanced Dataset								
Normal (312)	98.4%	97.1%	99.6%	99.6%	96.4%	93.2%	99.6%	99.6%
Keratoconus (312)								

Table 4
Confusion matrix of CMM-FFN.

Predicted class Actual class	Keratoconus	Normal
Keratoconus (<i>N</i> = 312)	291	21
Normal (<i>N</i> = 312)	1	311

Table 5
Confusion matrix of the CAD.

Predicted class Actual class	Keratoconus	Normal
Keratoconus (<i>N</i> = 312)	303	9
Normal (<i>N</i> = 312)	1	311

Table 6
Validation of The CAD system.

		10-Fold cross validation		
		KTC suspect vs. Normal	Mild KTC vs. Normal	Moderate KTC vs. Normal
CAD	Accuracy	96.56%	99.40%	99.58%
	Sensitivity	97.78%	98.81%	99.91%
	Specificity	95.56%	99.71%	99.90%
	Precision	95.65%	99.71%	99.90%
	Cut-Off	0.82	1	1
FFN-CMM	Accuracy	93.67%	99.55%	99.72%
	Sensitivity	90.00%	99.55%	99.44%
	Specificity	94.35%	99.56%	99.90%
	Precision	94.27%	99.56%	99.90%
	Cut-Off	0.82	1	1
BADD	Accuracy	89.00%	99.90%	99.02%
	Sensitivity	83.00%	99.85%	99.90%
	Specificity	95.00%	97.48%	98.05%
	Precision	95.00%	97.57%	98.15%
	Cut-Off	0.68	0.98	1
TKC	Accuracy	79.00%	93.11%	99.99%
	Sensitivity	58.00%	86.22%	99.99%
	Specificity	99.70%	99.99%	99.99%
	Precision	99.62%	99.99%	99.99%
	Cut-Off	0	1	1

results for mild and moderate KTC comparing with custom made mathematical model combined with feedforward neural network. However, the CAD is also able to successfully identify the earliest cases of keratoconus that cannot be detected clinically by looking at tomography maps alone, with an accuracy of 96.56% in detecting suspect KTC versus 93.67% for CMM-FFN (Table 6). In addition, the proposed approach outperformed Belin/Ambrosio Deviation (BADD) and Topographical Keratoconus Classification (TKC) systems in terms of accuracy, sensitivity, specificity, and precision for suspect KTC detection,

however, CAD shows nearly similar results in detecting mild and moderate stages in comparison with the previously described methods (Table 6). The high performance of CAD is also seen in the comparative ROC curves for the classification of keratoconus versus normal and suspect keratoconus versus normal (Fig. 5). Consequently, the algorithm accurately distinguishes keratoconus corneas from normal corneas at different stages.

3.2.3. Computational efficiency

The computational efficiency of the standard FFN, CMM-FFN and CAD is compared based on keratoconus groups (section 3.2.1) and using Holdout validation technique. The number of iterations, execution time and CPU time were selected from the most representative values, while training at least 20 times. Obtained results showed that the proposed algorithm overcomes the data complexities and reduces the computation time by 70% (Table 7). The validation performance of FFN-CMM converges after 17 iterations (epochs), considerably faster and with a considerably less variable error than standard FFN (Figs. 6 and 7). The stability and the convergence of FFN-CMM and CAD was investigated and presented in (Fig. 8) and (Table 7). CAD showed better convergence and less error variations, proving the distinct advantage of adding the Grossberg-RK2 layer to the standard algorithm.

The CAD showed an improvement in accuracy compared with the standard feedforward network (Table 7), that reached an accuracy of 98.4% vs 92.0% and a sensitivity of 97.1% vs. 83.7%.

4. Discussion

Many different modelling approaches have been suggested to assist practitioners with differentiating between normal and keratoconic corneas based on mathematical models [41,47,48,53–56] or machine learning algorithms. Other approaches consist of a direct statistical analysis of clinical quantitative parameters, such as corneal tomography, corneal topography or corneal pachymetry [10,28,29,32,33,68–72]. Most previous publications were platform dependent and rely on parameters that are often not provided by other devices, with very few exceptions [11,30,31]. This study presents a computer aided diagnosis system (CAD), based only on the elevation data and minimum corneal thickness, that can be captured by various ophthalmic devices. This might increase the interest and the potential reproducibility of the study. To the best of our knowledge, this is the first elevation-pachymetry based adapted algorithm that does not require additional clinical parameters. Moreover, the proposed approach is the first to consider the instability of machine learning algorithm for the early keratoconus detection.

Many earlier studies were based on classic machine learning algorithms [9,12,12,24,24,25,25,38–40,44,73–78] that did not consider the practical limitations of dealing with large, unstructured datasets and cost. The proposed framework overcomes this by reducing the (141×141) elevation matrices to 14 structured and meaningful features for each subject (Fig. 1). Consequently, the CAD shows a better classification performance than the standard feedforward network and reduces the computation time by 70% (Table 7). The CAD system also includes

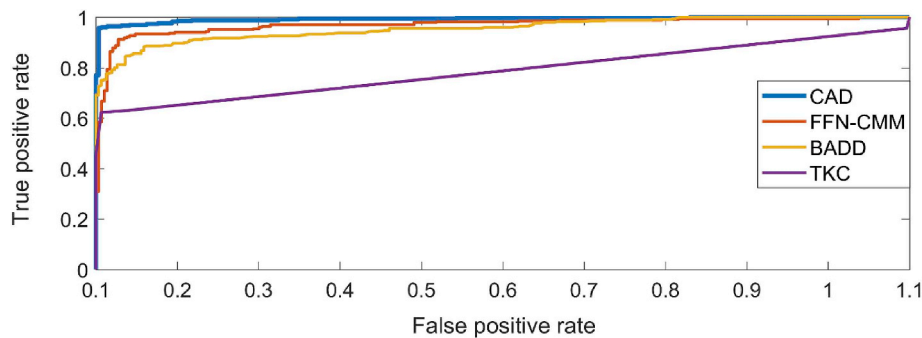


Fig. 5. Comparative ROC curves for the suspect KTC vs Normal.

Table 7

Comparison of the standard FFN, CMM- FFN, and CAD.

	FFN	FFN-CMM	CAD
Accuracy	92.2%	96.4%	98.4%
Sensitivity	83.7%	93.2%	97.1%
Specificity	99.0%	99.0%	99.6%
Precision	98.9%	99.6%	99.6%
Max. iterations	5	20	20
Number of features	117	14	14
Execution time	12.9 s	3.1 s	4.8 s
CPU time	248.0 s	20.0 s	6.1 s

an explicit Grossberg layer based Runge Kutta numerical schemes that increase the accuracy and stability of the standard feedforward neural network (Figs. 6–8), allowing the system to preserve the stability at different stages of the disease. Moreover, this work demonstrates that it is possible to improve keratoconus detection without using a large set of interdependent clinical parameters, and that the same approach can be used for detecting earlier stages of the condition (Table 6). The influence of the imbalanced data sets was investigated and validated using hold out validation technique (Tables 3–5), which demonstrated that the CAD is more stable and, more accurate results were reached by adding the Grossberg architecture.

A comparative study between the CAD and other available approaches for keratoconus detection at different stages was considered and validated using the 10-fold cross validation technique and ROC curves. This indicated that the CAD is indeed highly accurate in distinguishing clinical keratoconus from normal eyes with an accuracy of 99.58% (sensitivity: 99.91%; specificity: 99.90%; precision: 99.90%) for moderate cases and 99.40% (sensitivity: 98.81%; specificity: 99.71%; precision: 99.71%) for mild keratoconus. For these cases, the CAD provided nearly similar results as the feedforward combined with mathematical model (CMM-FFN), Belin/Ambrosio Deviation (BADD), and Topographical Keratoconus Classification (TKC) (Table 6). For the classification of keratoconus suspect, however, it improved on the performance of the other systems with an accuracy of 96.56%

(sensitivity: 97.78%; specificity: 95.56%; precision: 95.65%), compared to an accuracy of 93.67% for CMM-FFN (sensitivity: 90.00%; specificity: 94.35%; precision: 94.27%), 89.00% for BADD (sensitivity: 83.00%; specificity: 95.00%; precision: 95.00%), and an accuracy of 79.00% for TKC (sensitivity: 58.00%; specificity: 99.70%; precision: 99.62%). CAD can therefore accurately detects suspect keratoconus by dealing with the ambiguities found in the earliest cases of the disease and improving the convergence (Fig. 8).

In addition, the CPU time of the CAD was 6.1 s vs. 20.0 s using CMM-FFN and 248.0 s using only FFN, which demonstrate the computational efficiency of the proposed approach versus the traditional machine learning algorithms. Moreover, the achieved results showed the importance of custom-made mathematical model in reducing computational time and that the Grossberg-RK2 preserves the stability without causing an increase in computational cost.

Earlier work on detecting advanced KTC using Zernike polynomial fitting and decision trees reported an accuracy of 92.0% and an area under the ROC curve of 97.0% [39]. A follow-up study examined two orthogonal polynomials for spatial data transformation in conjunction with decision tree yielded to an easily interpretable results and less accuracy [40]. Sousa et al. [38] evaluated three machine learning classifiers for keratoconus detection, reporting a sensitivity of 99.0% with both support vector machine and multilayer perceptron against 98.0% with radial basis functions [38]. Another attempt to detect advanced keratoconus used discriminant functions to analyze corneal asymmetry parameters reaching a sensitivity of 94.0% and specificity of 100% [75]. More recent work analyzed tomographic, topographic and pachymetry data using support vector machine results an accuracy of 98.2% (sensitivity: 95.0%; specificity: 99.3%; precision: 97.9%) for keratoconus versus normal and an accuracy of 97.3% (sensitivity: 92.0%; specificity: 97.7%; precision: 78.8%) for early keratoconus [74]. While the latter study achieved very good results for keratoconus detection, their early keratoconus group includes keratoconus ectasia characteristics and the approach was based on a large number of datasets. The same limitation of ectatic characteristics was also encountered in another study [73] that used decision trees applied directly to 55 topographical parameters, and reported a sensitivity of

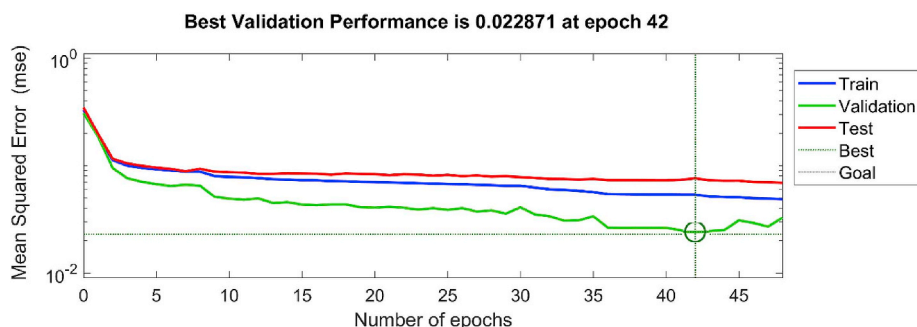


Fig. 6. Validation performance of the standard FFN.

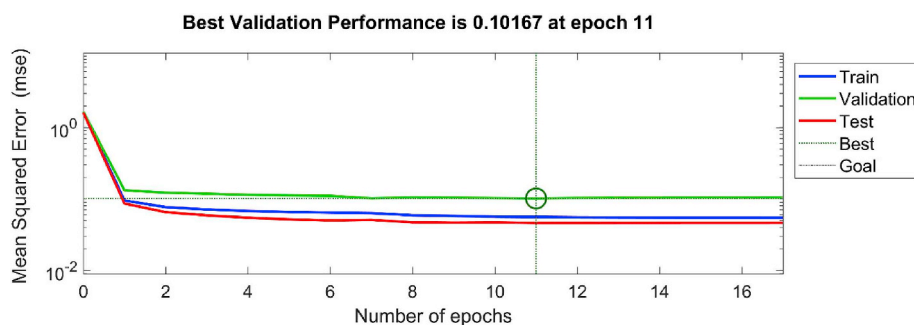


Fig. 7. Validation performance of the CMM-FFN.

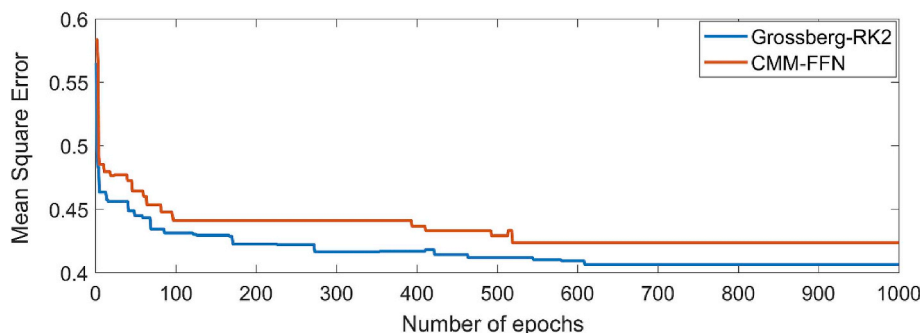


Fig. 8. Error changes with respect to the level of variations.

93.6% and specificity of 97.2% for distinguishing early from normal, and a sensitivity of 100% and specificity of 99.5% for advanced keratoconus. Meanwhile, other studies achieved a better sensitivity for earlier cases [73,74]. Similarly, our previous study using Support Vector Machine applied to a set of 22 Tomographical parameters from the Pentacam [22,23] reported an accuracy of 93.1%, sensitivity of 79.1% and specificity of 97.9% when classifying early KTC, and accuracy of 98.9%, sensitivity of 99.1% and specificity of 98.5% for advanced cases. Finally, Lopes et al. [77] examined tomographical data with various machine learning techniques, and achieved best results using random forests, reporting a sensitivity of 85.2% and specificity of 96.6% for suspect keratoconus. A more complete summary of studies that discriminate between normal and keratoconus was given in our previous study [23], and several reviews [79,80].

The proposed algorithm (CAD) is therefore comparable or higher to the earlier approaches for keratoconus detection. This is important since, even though recent studies report good results on earlier approaches, one often sees that their methodologies are based on proprietary parameters associated with one specific ophthalmological device. By modelling tomography data, which could be acquired from any commercially available Scheimpflug topographer, we overcame this issue while increasing the potential reproducibility of our study. Moreover, it deals with the problem of data complexity and the long training time (in the order of minutes or even hours for classical machine learning), while enhancing the detection accuracy for all stages of the disease. The proposed CAD deals with an important practical issue related to the instability of the algorithm while detecting ambiguous cases such as keratoconus suspect.

There are some limitations to this study that need to be acknowledged, most notably some false positive misclassifications that may be related to the clinical similarities between healthy and suspect stages of the disease (Tables 4–5). This issue can potentially be improved by adding other platform independent parameters. Another major limitation is related to the clinical definition of keratoconus suspect. Some authors define this group as corneas that show skewed radial axes as a first sign of KTC [76–78], while others considered it as the non-diseased cornea, while the fellow cornea is diagnosed with keratoconus

[22,24,50,81,82]. This work uses the latter definition since it is the most rigorous in the literature. However, using this strict definition comes at the price that some healthy corneas may be incorrectly considered as keratoconus suspect. This hinders direct comparisons with previous studies that used another definition, as it would lead to differences in accuracies, specificities, and sensitivities. Finally, the platform independence and the reproducibility of the proposed approach still need to be validated.

Further research could address additional issues, such as validating the CAD system using additional data collected at independent medical centers and using other devices. Another aspect to be explored is the ability of the algorithm to provide a scoring system for keratoconus progression, which forms another open and challenging question.

In conclusion, this study presented a computer aided diagnosis system that can accurately detect keratoconus at different stages. In principle this method should be platform independent, but this remains to be confirmed. Furthermore, the algorithm overcomes the data complexities issues, reduces the computation time and preserves the stability. The proposed framework presents an easy platform that can be used to assist ophthalmologist with the process of decision making. Finally, the main principles of the proposed strategy can be adapted to detect other diseases and yet reduces computation time.

Declaration of interests

None.

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